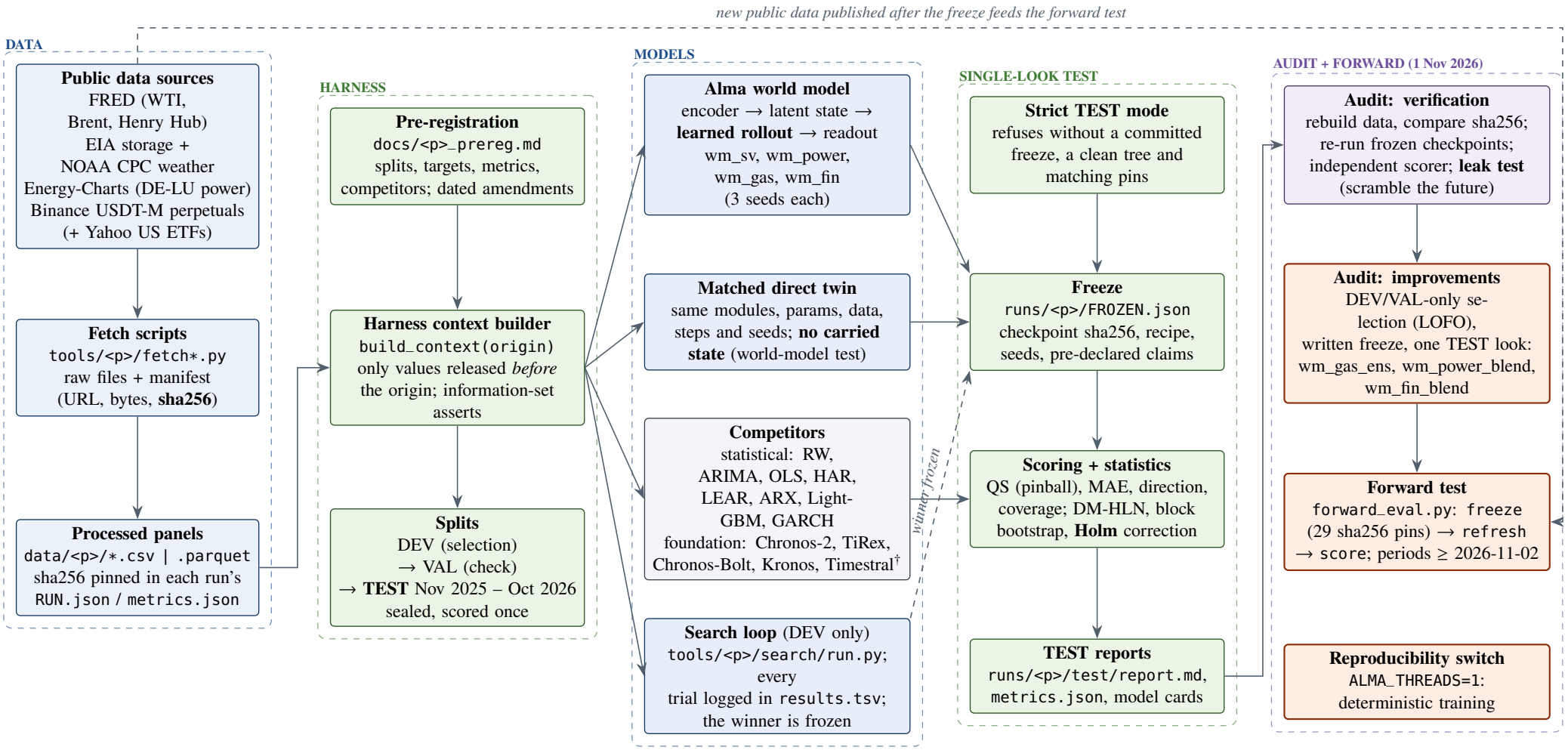
1. End-to-end system architecture

The pipeline every market project followed (oil, power, gas, crypto), with the audit and forward-test layers added on 1 November 2026. Source: `tools/<project>/`, `runs/<project>/`, `docs/*_prereg.md`, `tools/forward/`.



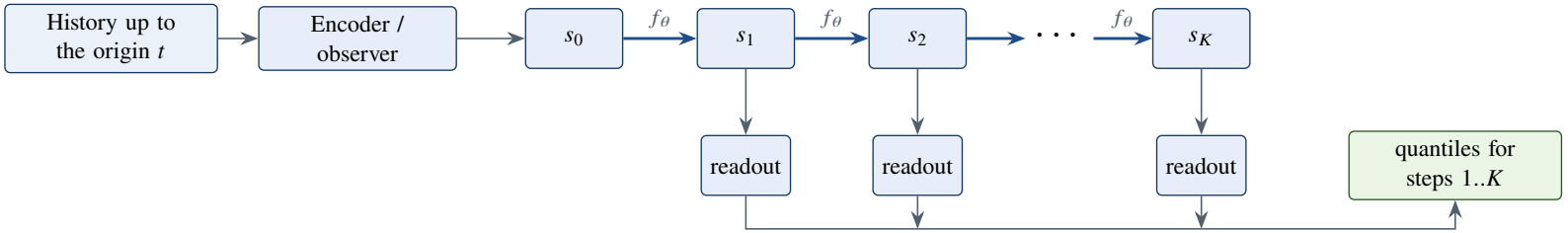
†Timestral-5M is scored but excluded from claims because its training data overlap the test period (flagged "contaminated" in every report).

data · original Alma model · added by the 1 Nov 2026 audit · evaluation · audit · baseline / other · latent rollout (carried state) · used in training only / optional

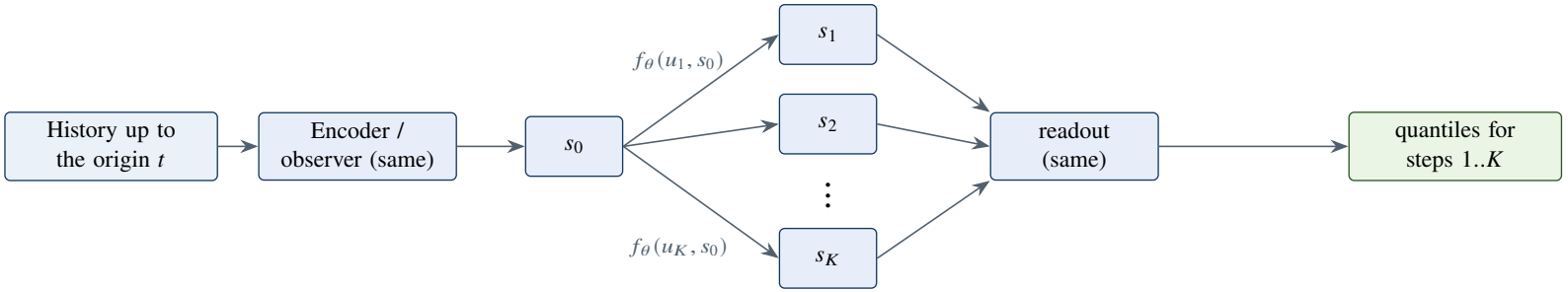
2. The core design: world model vs matched direct twin

Every Alma model has a twin with identical modules, parameter count, data, training steps and seeds. Only how the latent state advances differs. This comparison is the project's mandatory "world-model test" (`alma_goal.md` §2).

(a) World model (`arch=wm` / `abl=none`): $s_k = f_\theta(u_k, s_{k-1})$, the state is carried from step to step



(b) Direct twin (`arch=direct` / `abl=direct`): $s_k = f_\theta(u_k, s_0)$, every step is one move from the origin state



Pass rule (`alma_goal.md` §2)
The world model must beat its twin on the primary metric over ≥3 seeds, with a paired bootstrap CI that excludes 0.

Outcome on TEST
The twin is equal or better within noise in oil, power, gas and crypto.

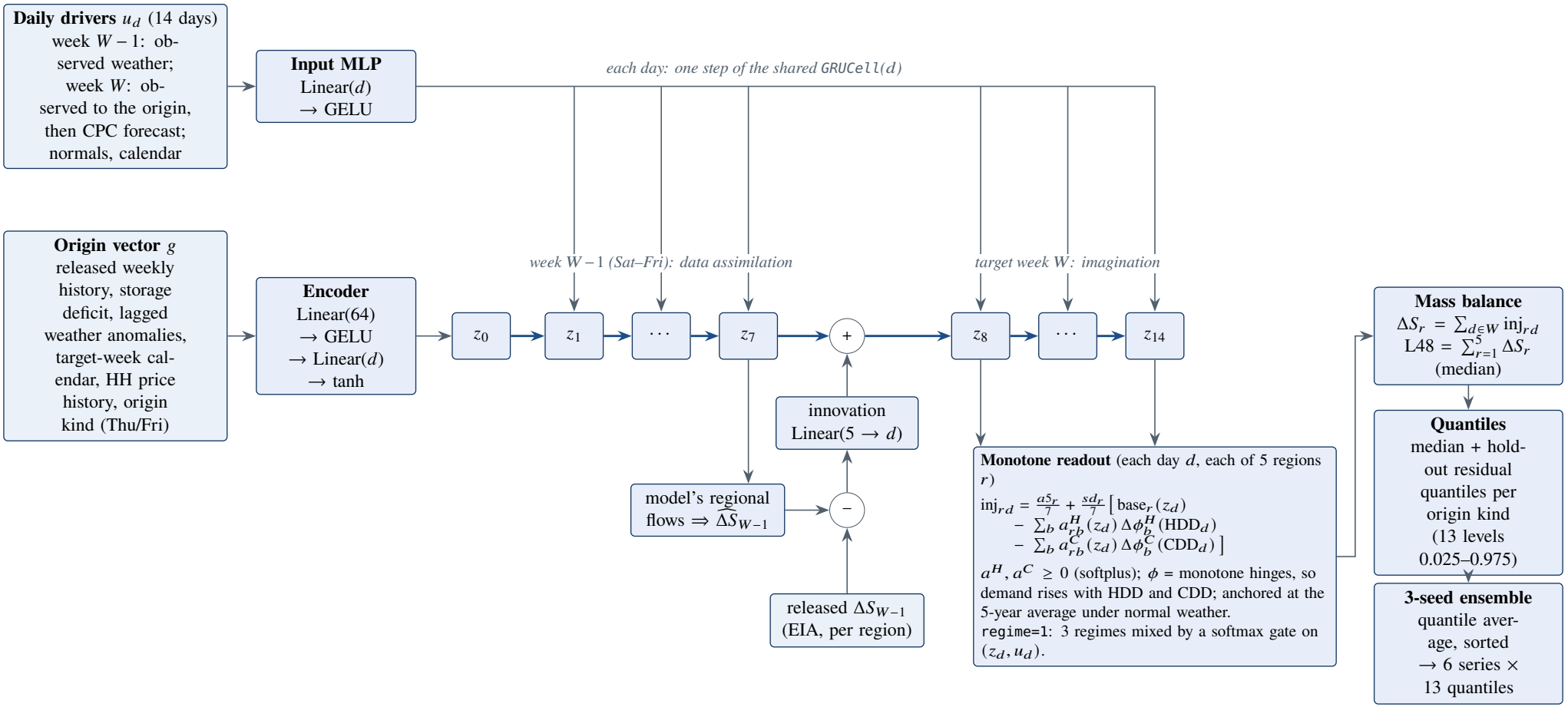
Research-track SSM, 5 seeds: rollout gain +0.7%, CI [−0.03%, +1.3%], not significant; +0.08% without the selection seed.

u_k is the step-k driver (calendar or phase clock, known or forecast exogenous inputs, lagged observations). Each project's twin is built by the same class with one flag changed: oil `SVWM(abl="direct")`, power `PhysWM(abl="direct")`, gas `GasNet(rollout=False)`, crypto `RSSM(abl="direct")`.

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3. wm_gas: storage-balance state-space world model (US natural-gas storage)

`tools/gas/search/nets.py` (`GasNet`), wrapper `tools/gas/wm_gas.py`. Frozen recipe: `arch=wm, assim=1, regime=1, d=32`, dropout 0.3, wd 0.05, 1,200 steps; seeds 0, 1, 2; trained on weeks released ≤ 2025-06-30.

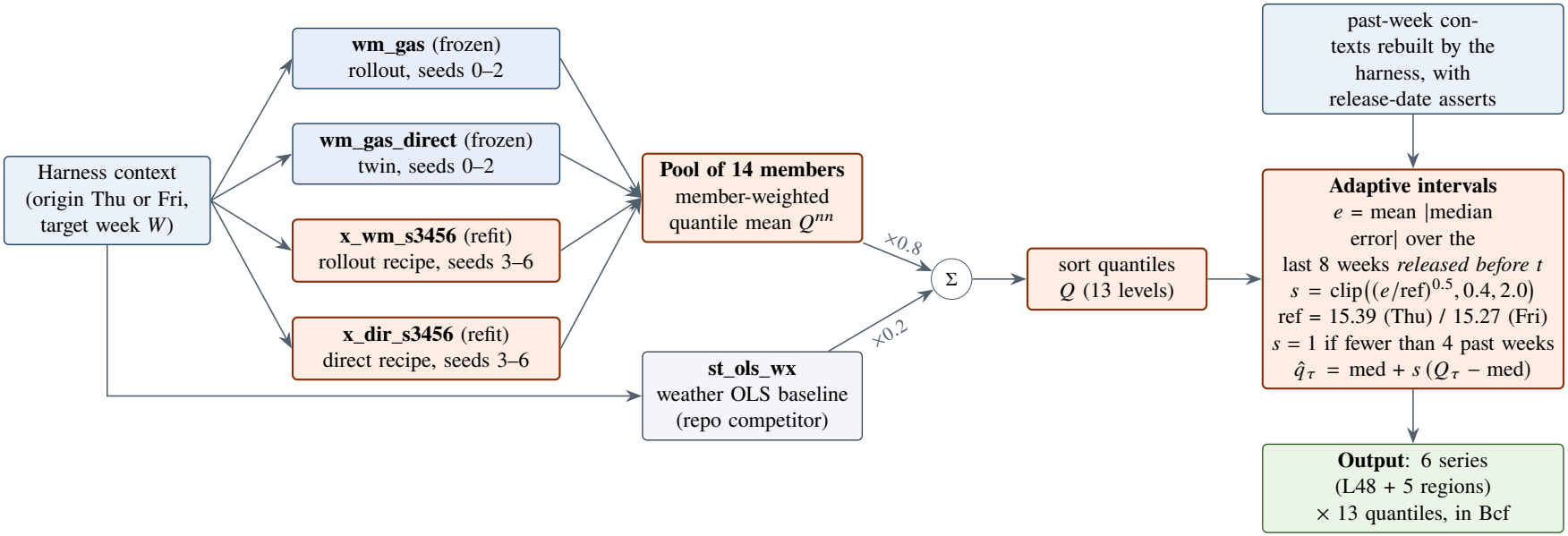


Twin (`wm_gas_direct`): each z_d is one GRUCell step from z_0 (no state carried between days). The innovation is computed in the same way, so both models see identical inputs. Size: about 23.5K parameters per member. **TEST** (Thursday, 52 weeks): MAE 11.53 Bcf, QS 4.59; the twin scores MAE 11.40, QS 4.57.

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4. wm_gas_ens: audit ensemble with OLS blend and adaptive intervals (new)

`tools/gas/wm_gas_ens.py`; checkpoints in `winning_models/gas_wm_gas_ens/` (sha256 asserted at load). Every choice was fixed on the DEV backtest (`audit/GAS_FREEZE.md`, written 16:34 and 17:00 UTC on 1 Nov 2026) before the TEST look.



How it was chosen (DEV, 6 rolling folds 2019-07 to 2025-06).

- The pool of rollout and direct seeds plus OLS, with the weight chosen leave-one-fold-out, reached DEV MAE 15.46 vs 16.23 for the frozen recipe with 7 seeds, lower in all 6 folds.
- Final frozen weight: 0.2 (full-DEV optimum); DEV MAE 15.39 / 15.27.
- Interval rule, chosen LOFO on DEV: DEV QS 5.970 → 5.919.

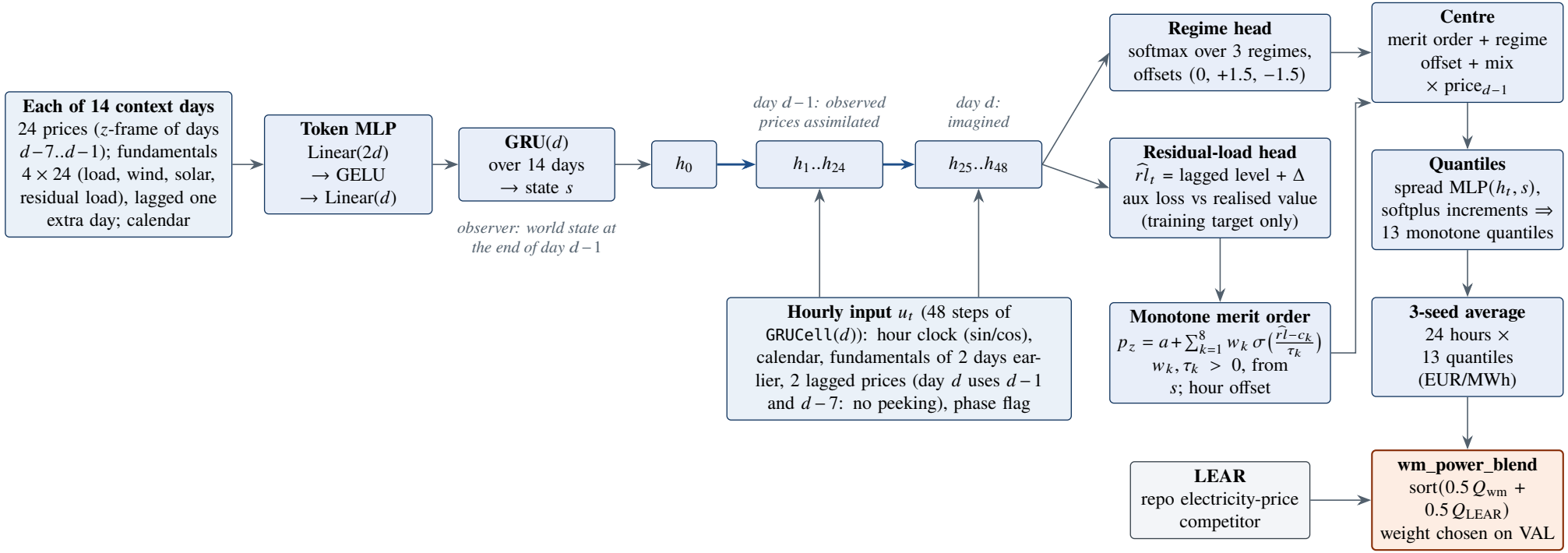
TEST result (52 weeks, Thursday / Friday).

- MAE 10.63 / 10.46 Bcf (wm_gas 11.53 / 11.56); QS 4.14 / 4.07 (4.59 / 4.56).
- Significant vs OLS on MAE and QS; not significant vs wm_gas.
- 80% coverage 0.885 / 0.904; intervals 17% narrower than with fixed widths.
- Leak test with the global data store scrambled: 0 of 22 forecasts changed.

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5. wm_power: physics-informed merit-order world model (German day-ahead power)

`tools/power/wm_models.py` (`PhysWM`, `arch=phys`), wrapper `tools/power/wm_power.py`. $d = 64$, 14 days of context, auxiliary loss weight 0.3, lr 10^{-3} OneCycle, batch 64, 1,500 steps, seeds 0–2. The blend was added by the audit.

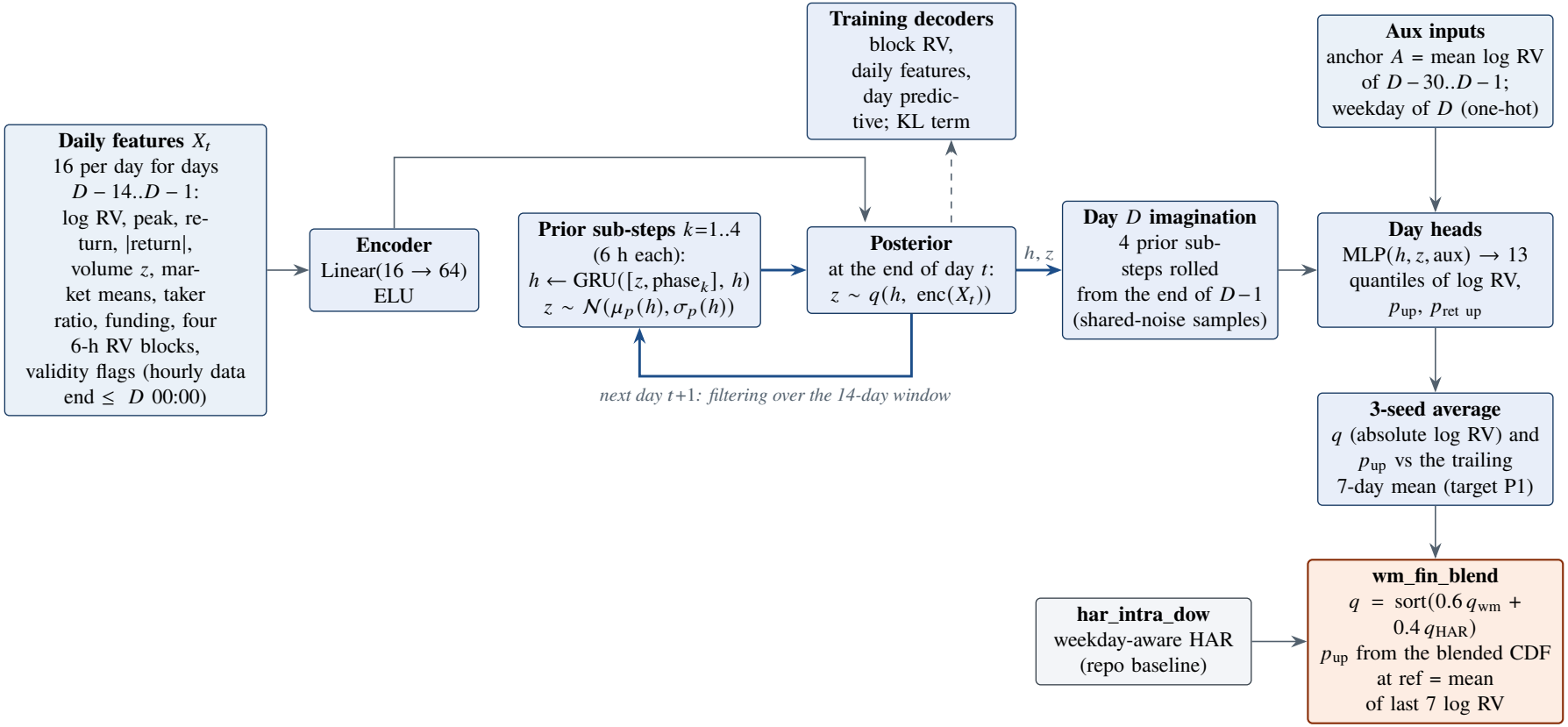


Twin (`wm_power_direct`): every hour is one GRUCell step from s. **TEST** (365 days): wm_power P2 0.747, QS 9.02, cov80 0.752; twin P2 0.739, QS 9.08. **Blend**: QS 8.60 (significant vs wm_power and LEAR), P2 0.755 (significant vs LEAR only).

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6. wm_fin: RSSM world model on an intraday clock (crypto realized volatility)

`tools/fin/wm_core.py` (`RSSM`, "DreamerV3-lite"), wrapper `tools/fin/wm_fin.py`. $d = 64$, $z = 16$, 4 prior sub-steps (6 h each) per day, window $W = 14$ days, weekday-of-D input; lr 2×10^{-3}, 800 steps; 3 seeds refit on labels $\leq$ 2025-06-30 (16 training coins). The blend was added by the audit.



Twin (`wm_direct`): the day heads read the filtered state at the end of $D-1$, with no day-D rollout (identical modules and losses).

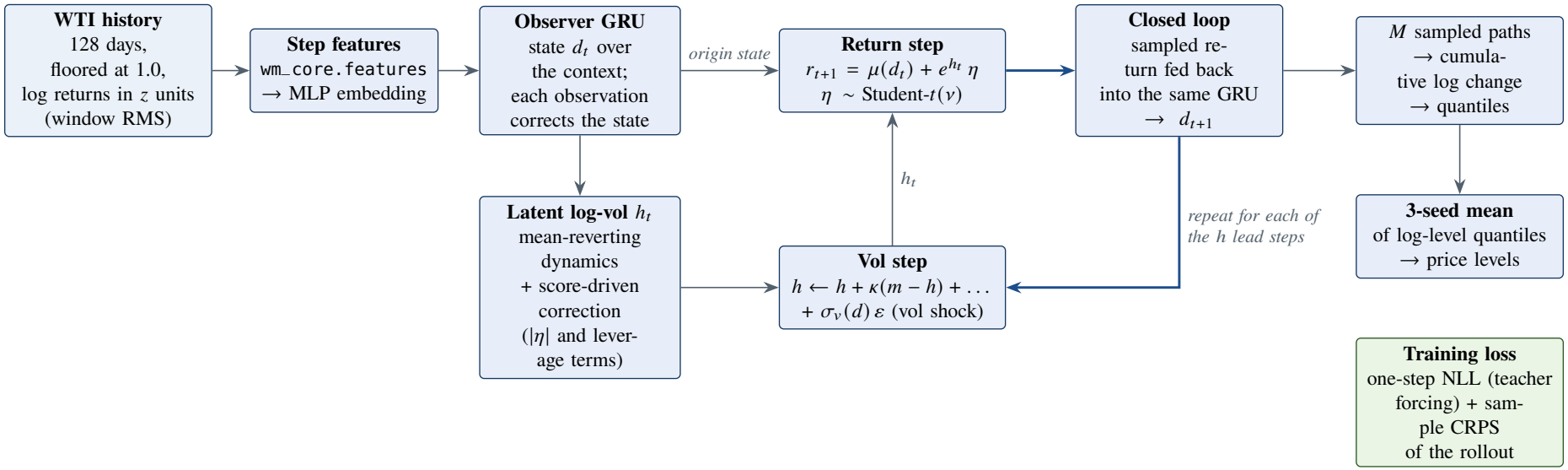
Other frozen pillars (not diagrammed): wm_fin_cl (continual SGD, 1 step per day), wm_agent (Dreamer-style actor-critic), PPO/DQN twins.

TEST (24 coins): wm_fin P1 0.727, QS 0.1471; twin P1 0.724, QS 0.1465; har_intra_dow P1 0.731. **Blend**: P1 0.740, QS 0.1459. Significant vs wm_fin on P1; not vs har_intra_dow.

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7. wm_sv: stochastic-volatility world model (WTI crude oil)

`tools/commodity/wm_models.py` (SVWM), wrapper `tools/commodity/wm_commodity.py`. Input: 128 days of WTI only. $d = 64$, about 25,191 parameters per member; lr 2×10^{-3} cosine, batch 64, 600 steps; training windows end before 2023-01-01; seeds 0–2, mean of log-level quantiles.

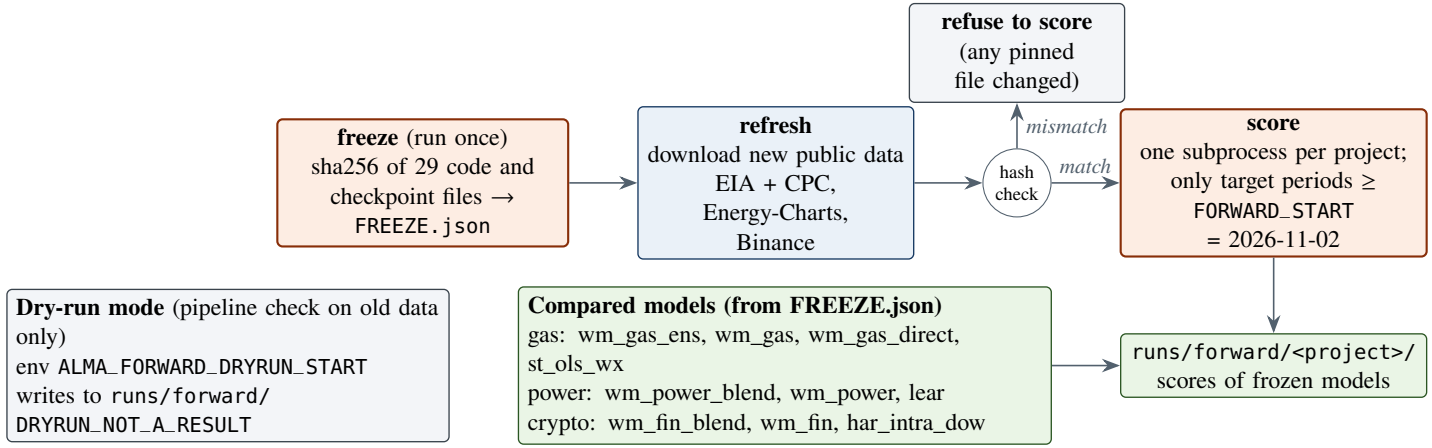


Twin (`wm_sv_direct`): lead k is one step from the origin state, $d_k = \mathrm{GRU}(\mathrm{emb}_k, d_{\mathrm{origin}})$, decoded as a Student-$t$ with scale $e^{h_{\mathrm{origin}} + m(d_k)}\sqrt{k}$; nothing is sampled step by step.
TEST ($h = 1$, 246 origins): QS 0.01052 vs random walk 0.01056 (skill +0.45%, $p = 0.479$, a tie); twin 0.01056. No improvement model was built, because the signal is indistinguishable from a random walk.

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8. Forward-test architecture (frozen 1 November 2026, 17:47 UTC)

`tools/forward/forward_eval.py`, `tools/forward/FREEZE.json`. Scores the frozen models only on data published after they were frozen.



Data arrival: first EIA storage report 2026-11-05; gas release #20 on 2027-03-18 and #52 on 2027-10-28; power and crypto day 60 on 2026-12-31 and day 365 on 2027-11-01. These follow the publication calendars, with holiday shifts possible.

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